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Design and implementation of real-time monitoring system for multiple machine tools

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Abstract

With the increase in the number of machine tools in the workshop and the increase in machine tool operating data, the traditional maintenance mode cannot meet the needs of enterprises for efficient and stable production in the workshop. Based on big data components such as Flink, Kafka, and HBase, a high-performance real-time monitoring system for multiple machine tools is constructed. The system collects data by adding sensors and the communication protocol of the CNC system, and writes the collected data into the message queue Kafka. The real-time monitoring module is built with Flink as the core, and consumes data from Kafka as a Kafka consumer. After Flink completes the preprocessing and statistics of the data, it stores the data in HBase, and realizes the visual display of the monitoring data through visual components. Experimental test results show that the system runs stably, has reliable performance, and has good scalability, which can effectively improve workshop monitoring efficiency and reduce enterprise operating costs.

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1. Introduction

Machine tools are machines for manufacturing machines and machinery, also known as industrial master machines, and are the foundation of modern industry. Usually, CNC companies often have multiple CNC machine tools, and these machine tools are often in a state of decentralized management. At present, there have been many researches on the monitoring of machine tool operation status. For example, Qin Meng¹ proposed a design scheme

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of a remote monitoring system for CNC machine tools based on the MQTT protocol and Web framework; Sun Shunmiao² proposed a design based on Storm stream processing Monitoring method of machine tool operation status. However, most of the existing researches on machine tool monitoring systems are limited to the monitoring of the operating status of a single machine tool. The operator needs to check on-site in real time to learn the operating status of the equipment, which is inefficient. As the company further expands its production scale and purchases more machine tools, the workload of operators also increases. Therefore, it is of great significance to design a monitoring system that can accurately and real-time reflect the running status of multiple machine tools.

2. Design and implementation of monitoring system

The machine tool monitoring system consists of an acquisition module, a buffer module, a real-time calculation module, and a storage and interaction module, as shown in Fig. 1.

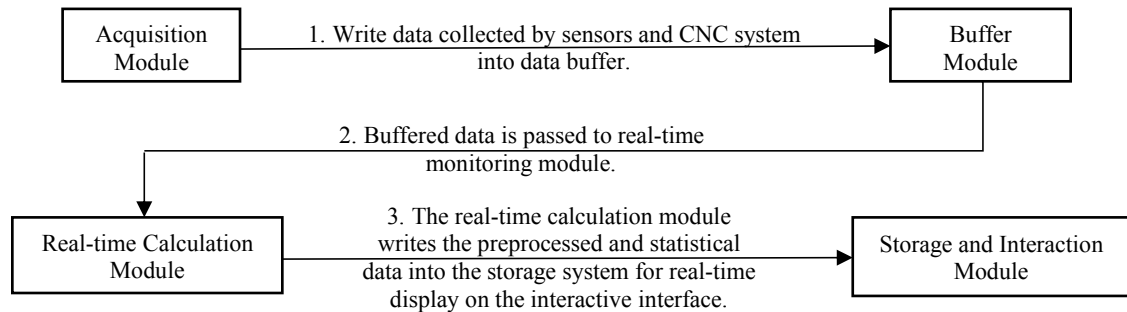


Fig. 1. Real-time monitoring system architecture

2.1. Acquisition Module

Due to the large number of machine tool components, there are different collection methods for the working status of each component. Generally speaking, it can be divided into two: the collection method based on the communication protocol of the CNC system and the collection method with additional sensors.

2.1.1. Acquisition based on CNC system communication protocol

At present, many CNC systems provide external communication protocols. For example, common foreign Siemens and Fanuc CNC systems, domestic Huazhong CNC and Guangzhou CNC, etc., all meet the OPC specifications and are equipped with corresponding Ethernet interfaces. CNC machine tools with Ethernet interfaces can perform data through OPC specifications. Communication. As long as the communication connection between the computer and the numerical control system is established, the operating status of the machine tool, the position and speed of each axis and other information can be collected³.

2.1.2. State data collection based on external sensors

The state information of the mechanical parts of the CNC machine tool body, such as the spindle, tool holder, etc. can be collected by installing sensors on the machine tool parts to improve the data acquisition system⁴. This article mainly installs the temperature sensor and acceleration sensor on the machine tool body to collect the temperature and acceleration signals of the machine tool.

2.2. Buffer Module

Due to the large amount of data collected by the state of the machine tool and the fast speed, the actual production environment may not be synchronized with the processing speed of the real-time computing module, so a data buffer queue needs to be introduced to coordinate the data transmission.

Kafka is a distributed, publish and subscribe messaging system. Its use offset ensures that messages can be consumed in an orderly and unique manner. By introducing consumer groups, a set of data can be consumed by multiple consumers at the same time, which increases the throughput of the system. At the same time, Kafka also keeps multiple copies of the data in each partition to achieve high data availability. Even if one machine fails, the data in Kafka will not be lost in most cases⁵. The principle of Kafka is shown in Fig. 2.

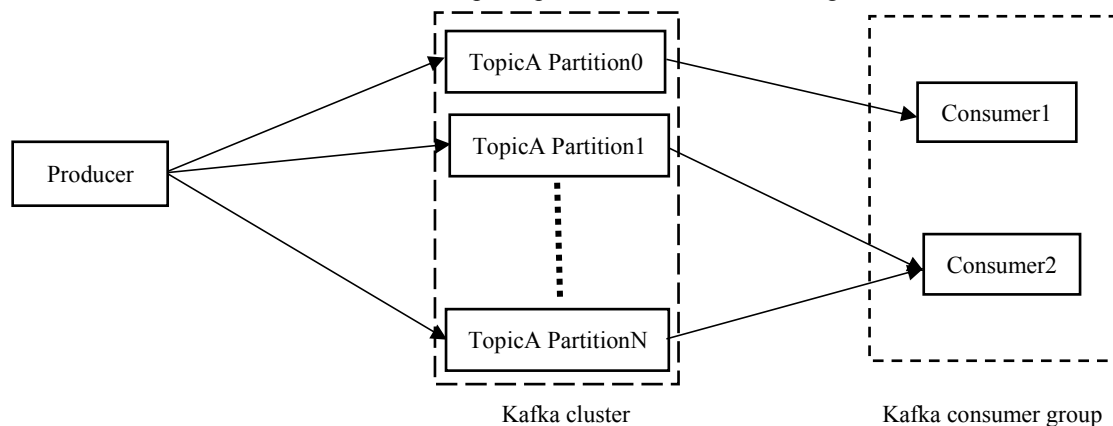


Fig. 2. Schematic diagram of Kafka principle.

This article regards the acquisition module as Kafka's Producer and the real-time computing module as Kafka's Consumer to ensure real-time and stable data transmission.

2.3. Real-time Calculation Module

Streaming data is an unbounded, continuously arriving data set. The processing of streaming data requires high real-time computing power and continuous computing power⁶. Commonly used stream computing engines include Spark Streaming, Storm, S4, etc.⁷. Considering that Flink has the characteristics of millisecond delay, stateful calculation, and rich time semantics, this paper selects Flink as the stream processing engine of the machine tool monitoring system, and the operating principle⁸ is shown in Fig. 3.

Flink is mainly composed of three components in operation, JobClient, JobManager and TaskManager. The user first submits the Flink program to the JobClient, and submits it to the JobManager after processing, analysis, and optimization by the JobClient, and finally the task is run by the TaskManager.

Due to electromagnetic, noise and other factors existing in the actual production environment may interfere with the collection of the sensor, resulting in obvious abnormal values in the actual collected data. In addition, the sensor may malfunction or even stop working, which will also affect the quality of the collected data. Therefore, data preprocessing must be performed before the machine tool status is formally judged. In this paper, outliers, null values, and redundant data are filtered in the preprocessing.

Since there may be status information of multiple machine tools in Kafka, it is necessary to perform keyBy operations on the data stream before judging the running status of each machine tool to obtain the KeyedStream of each machine tool, and judge the running status of each machine tool in the KeyedStream.

For the judgment of the operating state of the machine tool, an algorithm model can be trained for each machine tool component separately⁹, and then through the model and the collected data to determine whether the component is in a normal operating state. This paper adopts the method of pre-setting a threshold for each monitoring index [10], and an alarm is issued when the real-time monitoring value of a monitoring index exceeds the threshold. The specific implementation method is:

- 1) Create a table in MySQL to store the alarm thresholds of various monitoring indicators.

- 2) The TimerService in Flink's Context object provides the registerEventTimeTimer method, which accepts a timestamp and triggers the timer task when the event time exceeds the timestamp.
- 3) The ProcessFunction interface of Flink provides the onTimer callback function, which will be called when the timer set in 2) is triggered.
- 4) Read the alarm threshold set in 1) in the callback function, and update the timestamp in 2) to make it 5 seconds later than the time of the current event.
- 5) After 5 seconds, the timer will be triggered again, repeating the cycle from 2) to 5).

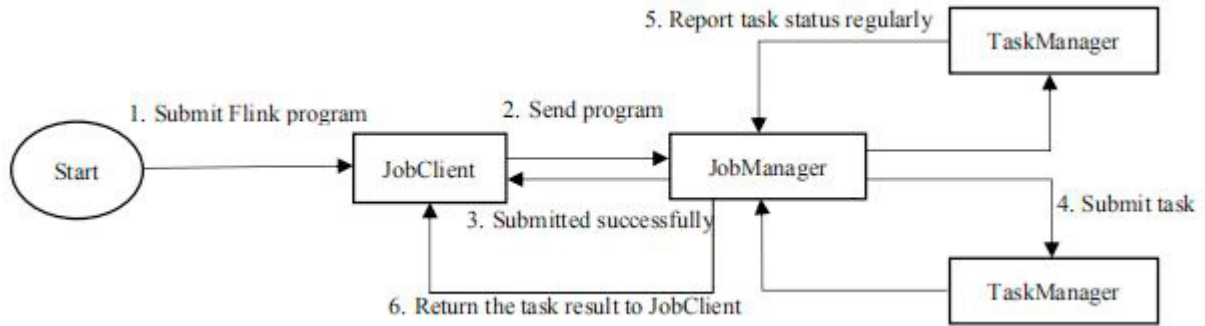


Fig. 3. Flink operating principle diagram.

The user does not need to close the program to modify the code, just modify the threshold set in MySQL through the interactive page to change the alarm threshold of the monitoring program. The processing flow of the real-time computing module is shown in Fig. 4.

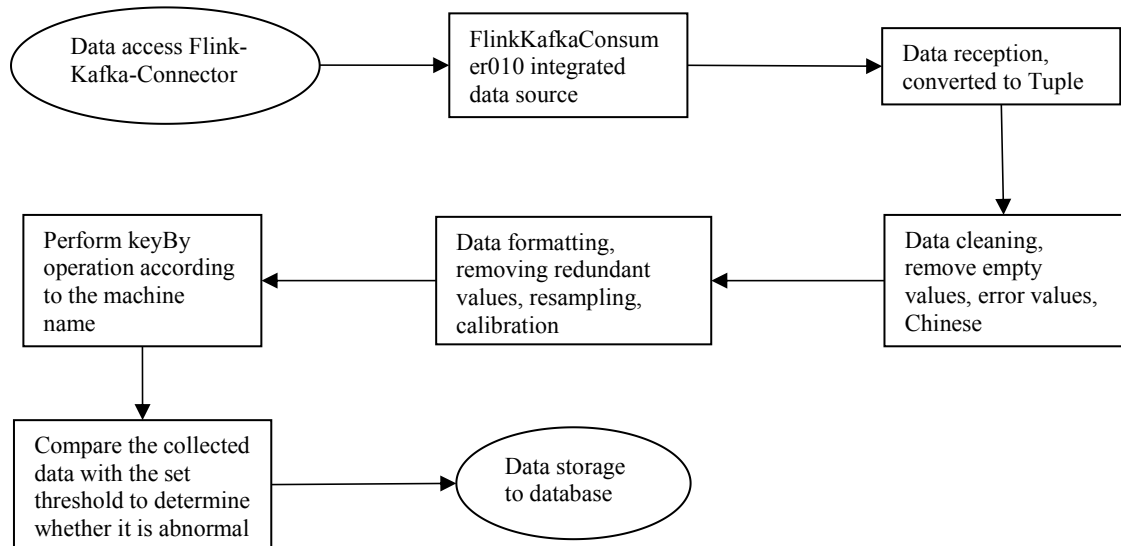


Fig. 4. Real-time computing module processing flowchart

2.4. Storage and interaction module

The monitoring system needs to store two kinds of data, one is the status data of the machine tool running process, this kind of data has the characteristics of massive, low frequency of use, etc.; the second is user information, such as user name, password, etc., and the machine tool set by the user Alarm threshold, these data are usually small in

scale, but the frequency of query and modification is high. HBase is a distributed NoSQL database that supports real-time reading and writing and random access to very large data sets¹¹. Although there have been many strategies and implementation methods for data storage and access, in fact, most solutions, especially some relational ones, were constructed without considering the characteristics of super-large scale and distributed. Therefore, when designing a storage system, the former is stored in HBase, while the latter uses traditional relational databases, such as MySQL.

MONITORING SYSTEM

Username

Password

Verify Code


Sign up

Reset

Fig. 5. User login interface

The user interaction module adopts the B/S architecture and uses the visualization component ECharts to construct a browser-based user interaction page, which mainly includes user login, machine tool management and real-time monitoring interfaces. The user login interface is mainly used to verify the identity of the logged-in user. To protect user information, after entering the password on the login page, the corresponding input character should be replaced with "." or "*" using SetPasswordChar, as shown in Fig. 5. The machine tool management interface can add and cancel the monitoring of machine tools, and set alarm thresholds, as shown in Fig. 6. The real-time monitoring interface can observe the operation details of all machine tools, as shown in Fig. 7.

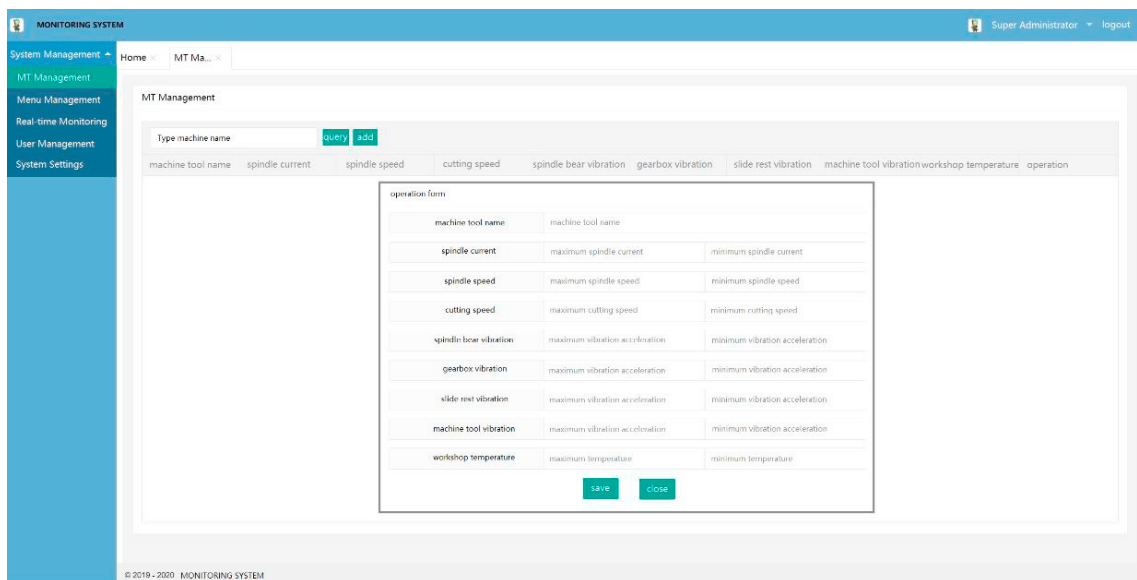


Fig. 6. Machine tool management interface

3. Design and implementation of monitoring system

In this paper, the machine tool of Siemens 840D CNC system is used as the research object. The vibration sensor of model NX-0901 is installed on the spindle bearing of the machine tool, the central gearbox of the spindle, the turning tool post and the outside of the machine tool; Temperature Sensor. At the same time, the OPC client is developed to access the OPC server built into the CNC system to collect status information such as the actual current of the spindle, the actual speed of the spindle, the current cutting speed, and the current position of the machine tool.

Install Flink, Kafka, HBase and other open source software on 8 PCs with the same configuration to build a distributed cluster environment. Count the processing time of the program when monitoring different numbers of machine tools under the fixed acquisition frequency of 1600HZ, and the result is shown in Fig. 8.

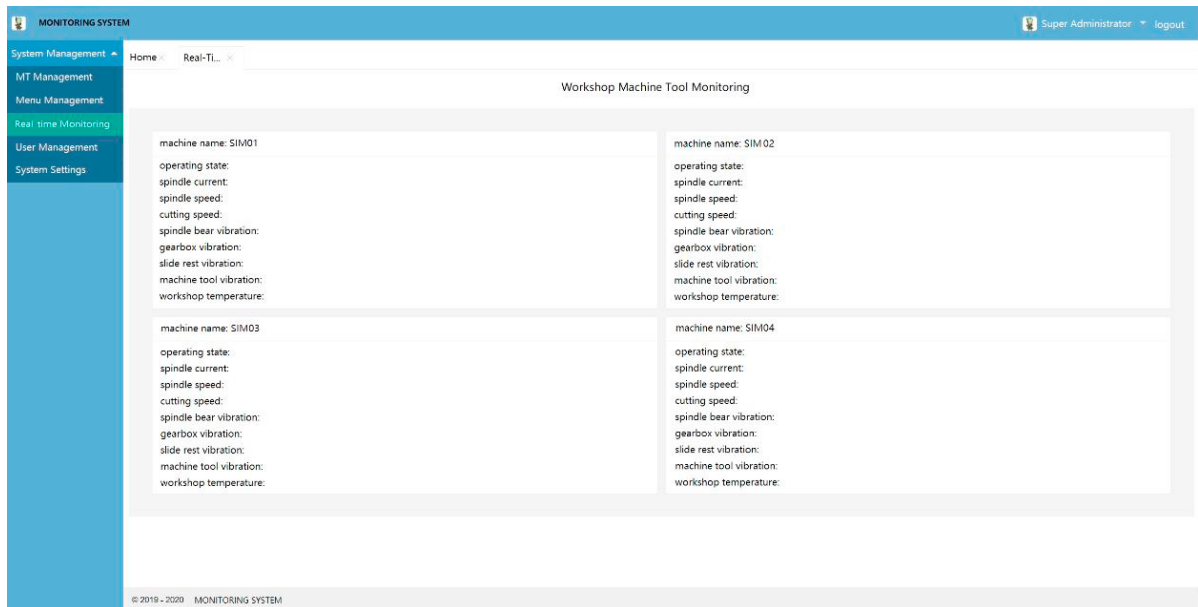


Fig. 7. Real-time monitoring interface

It can be found from Figure 8 that due to the continuous increase in the number of machine tools, the amount of data processed by the monitoring system has increased sharply, and the execution time of the program has also increased, but it still remains in a relatively short time. As the number of machine tools further increases, the monitoring system needs to process more data. At this time, the performance of the monitoring system can be improved by increasing the number of machines in the Flink cluster.

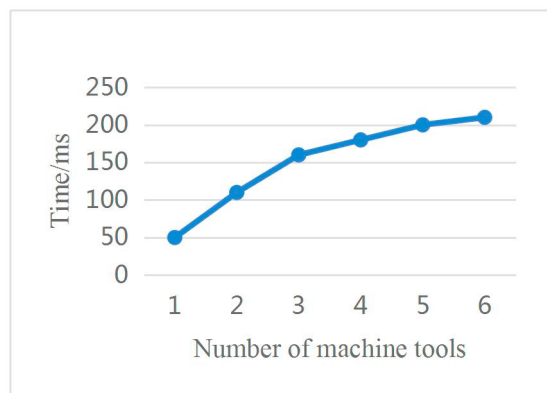


Fig. 8. Comparison of the number and execution time of different machines

4. Concluding remarks

Based on Flink, Kafka, HBase and other big data components, this paper uses the CNC machine tool of Siemens 840D CNC system as the experimental platform to build a reliable and easy-to-expandable real-time monitoring system. The implementation of the monitoring system is described in detail in terms of the collection, buffering, real-time calculation, storage, and visualization of the operating status of the machine tool. The monitoring system is deployed in the actual production environment for experiments. The experimental results show that the system has strong real-time processing capability for the monitoring of multiple machine tools in the workshop, and the system runs reliably.

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